MIL-D-18163A(SHIPS)
31 January 1962
SUPERSEDING
MIL-D-18163(SHIPS)
20 September 1954

MILITARY SPECIFICATION

DRY-CLEANING UNIT ASSEMBLY, FULLY AUTOMATIC, SYNTHETIC SOLVENT (PERCHLORETHYLENE) TYPE, (25-35 POUNDS CAPACITY) SHIPBOARD USE

1. SCOPE

1.1 This specification covers fully automatic dry-cleaning machines of the closed system type, for using perchlorethylene solvent, for Navy shipboard service.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose)

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of
MIL-P-15137 - Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use)
MIL-B-16392 - Brakes, Electric, Naval Shipboard

FSC 3510

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MILITARY (cont'd)

MIL-T-16420 - Tube, 70-30 and 90-10 Copper-Nickel Alloy, Seamless and Welded
MIL-B-16541 - Bronze, Valve, Castings
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

PUBLICATIONS

BUREAU OF SHIPS
NAVSHIPS 4786 - Recommendation and Proposals for Onboard Repair Parts

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

AMERICAN INSTITUTE OF ELECTRICAL ENGINEERS (AIEE)

45- Recommended Practice for Electrical Installations on Shipboard.

(Application for copies should be addressed to the American Institute of Electrical Engineers, 33 West Thirty-ninth Street, New York 18, New York.)

NATIONAL BOARD OF FIRE UNDERWRITERS
National Electrical Code.

(Applications for copies should be addressed to the National Board of Fire Underwriters, 85 John Street, New York 7, N. Y.; 222 West Adams Street, Chicago 6, Illinois, or Merchants Exchange Building, San Francisco 4, California.)

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OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Preproduction.- Prior to the start of production one unit of each design and size shall be submitted for the preproduction tests specified in 4.2 to determine whether or not the dry-cleaning unit is suitable for shipboard application.

3.2 Commercial operation. - No machinery or equipment will be acceptable unless dry-cleaning units of approximately the same type and design as that specified herein have been operating successfully in a commercial dry cleaning plant or plants for at least one year. Units installed for test purposes in a manufacturer's plant or laboratories shall not come within the category of successful commercial operation.

3.3 Materials.- The materials used for construction of the frames, base, and strength members shall be malleable iron, ductile iron or steel, or materials regularly used by the supplier. Gray cast iron or other brittle materials shall not be used for these parts, except as otherwise specified herein. Gray cast iron may be used for other components, providing a unit so equipped, satisfactorily completes the vibration and impact test procedures outlined herein.

3.4 Design and construction. -

3.4.1 General requirements. - The design, materials of construction, fabrication, finish, operation, and details of component assemblies shall be suitable and adequate for the purpose intended. The interior and exterior of the units shall be thoroughly cleaned and finished. All surfaces with which clothes come in contact shall be smooth and free from sharp edges and projections. Tolerances shall be in accordance with good commercial standards.

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3.4.1.1 The dry-cleaning unit shall be of rigid construction and quiet in operation and shall be designed to provide for manual control. All bearings shall be readily replaceable. All water cooled equipment shall be designed for using sea water.

3.4.1.2 The complete unit shall be new and furnished complete with motors, driving mechanism, starters, and controllers, including master switches, timers, reversing mechanisms, if used and other electrical equipment, if and as applicable, with wiring installed in rigid metal conduit or electric metal tubing (except that flexible conduit may be used as necessary to permit adjustments of motors which drive belts) and permanently connected.

3.4.2 Inclined operation. - The dry-cleaning units shall function in accordance with the requirements of this specification without reduction in performance when the ship is rolling 30 degrees starboard in a 4-1/2 second period.

3.4.3 Overall dimensions. - The dry-cleaning unit shall be compact as practicable. However, all components shall be accessible for inspection and maintenance. The overall length of the unit shall not exceed 12 feet. The height shall not exceed 7 feet. The depth shall not exceed 6 feet. However, the total deck or floor space required by the dry-cleaning plant, exclusive of the press and spotting board, shall not exceed 45 square feet.

3.4.4 Drives. - The shafts, bearings, gears and similar items shall be designed so that the maximum stresses in all parts do not exceed 70 percent of the yield strength of the materials used under maximum possible applied operating torque and under maximum full load operating conditions.

3.4.5 Gearing, belts, chains. - All gearing, belts, chains, (if used), and shafting shall be selected as recommended by the manufacturer of the component.

3.4.6 V-belt drives. - V-belt drives shall be selected as recommended by the belt manufacturer with load ratings, including applicable safety factors, greater than the maximum belt loads under extreme operating conditions. The V-belts shall be capable of a minimum service life of 5,000 hours of operation of the unit.

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3.4.7 Metal joints. - Rivets, welds, bolts, screws, nuts, and washers shall be of steel, except when brass or corrosion-resisting metal is fastened, in which case they shall be of brass or corrosion-resisting metal. Where dissimilar metals are fastened, welds, bolts, rivets, screws, and nuts shall be of the higher grade metal. Electrodes or welding rods used for welding shall deposit material similar to the base metal. Nuts subject to shock or vibration shall be adequately locked.

3.4.8 Piping. - All interconnecting pipes, fittings, and valves within the dry-cleaning unit, shall be installed on the units by the contractor. All valves shall be readily accessible for operation and maintenance.

3.4.8.1 The pipes, fittings, and valves shall be of the following material:

Solvent - - - - -	Galvanized iron.
Condensate - - -	Galvanized iron.
Steam - - - - - -	Black iron.
Air - - - - - - -	Copper.
Salt water - - - -	Pipes and fittings: Copper-nickel-alloy, MIL-T-16420. Valves: Bronze housing in accordance with MIL-B-16541, using nickel-copper-alloy (minimum 60 percent nickel) for internal working parts.

3.4.8.2 All piping shall be designed for an operating pressure of 100 pounds per square inch (p.s.i.).

3.4.9 Bearings. - Bearings shall be selected in accordance with FF-B-171 as recommended by the bearing manufacturer with load ratings, including applicable safety factors, greater than the maximum bearing loads under extreme operating conditions.

3.4.10 Solvent. - The equipment shall be designed for using synthetic solvent (perchlorethylene).

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3.4.11 Electrical conduit and wiring. - Electrical conduit shall be oil resistant, flexible or rigid steel, zinc-coated. All wire shall be stranded, insulated, heat-resistant grade. Conduit, fittings, and wire shall confrom to and be installed in accordance with the National Electric Code. Equipment will be used on an ungrounded electrical system and no grounds shall be permitted on the machine wiring.

3.4.11.1 Wiring and instructions. - All connections shall be made between motors, controllers, timers, pushbuttons, and limit switches installed on the dry-cleaning units. Motor circuit wiring shall be not less than No. 12 AW gage, larger sizes being used where required by service requirements. Wiring for pilot control circuits to pushbutton stations, limit switches, shall be not less then No. 14 AW gage. When magnetic controllers are remotely mounted separately from the unit, all leads shall be extended from motors and control devices to a connection box and properly identified for connections to wiring beyond the dry-cleaning units. Instructions for connecting leads shall be furnished with the unit. Machines with manual across-the-line starters when mounted on the unit shall be provided with wiring between the motor and starter.

3.4.12 Motors. - Motors shall have ample power to operate units for which they are designed, under full load operating conditions, without exceeding permissible temperature rise. Reversible motors shall be suitable for the required frequency of reversals. Motor bearings shall have adequate facilities for lubrication. Motors shall have proper starting and operating characteristics for the driven equipment. Motors shall be either a.c. or d.c. voltage, as specified (see 6.1).

3.4.12.1 Motors, motor controllers and associated pushbuttons, switches and indicator lights shall conform to the requirements of AIEE 45.

3.4.13 Dry-cleaning unit. - The dry-cleaning unit shall be a self-contained integral unit, complete with combination washer-extractor-dryer, deodorizer, still, condenser, filter, storage tank, pumps solvent gauges, and all necessary pipes, valves, and fittings, and shall be fully automatic of the closed system, straight solvent, recovery type having an operating cycle of 30 to 35 minutes. The dry-cleaning units shall be the front loading type (defined as one having its cylinder axis of rotation perpendicular to the front of the machine).

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3.4.13.1 Deodorizing exhaust.- The unit shall include an exhaust fan which will function during the deodorizing phase of the operation. The fan shall exhaust not less than 400 c.f.m. overboard through a suitable duct 100 feet in length with two 90-degree bends. The air velocity through the duct shall not be less than 2500 feet per minute (f.p.m.).

3.4.13.1.1 Duct. - The duct, external to the unit shall not be furnished by the contractor for the unit.

3.4.13.2 Size and capacity. - The dry-weight load capacity of the dry-cleaning units shall be based upon the cubical capacity of the cylinder in inches divided by the approximate required volume (cubic inch) per pound of clothes which varies with the diameter of the cylinder as specified (see 6.1):

Cylinder diameter Inches	Capacity Cubic inch per pound dry weight
24	660
30	600
36	550

3.4.13.2.1 Dry-cleaning units having 2-pocket cylinders shall be considered to have 15 percent less dry weight load capacity than units with single-pocket cylinders.

3.4.13.3 Shell. - The shell shall be of steel construction with plastic, zinc or lead interior coating. The shell shall be solvent-tight and of sufficient thickness and rigidity to prevent distortion or objectionable vibration when the unit is in operation. The joints between the upper and lower halves shall be solvent-tight. The shell plates shall be continuous in the lower half of the shell. The shell shall be fitted with a suitable size door and shall be constructed to provide for the removal of the cylinder.

3.4.13.3.1 Shell doors. - The shell shall be fitted with a reinforced hinged or sliding door. Shell doors shall latch in both the open and closed positions. The shell door shall be designed for solvent-tight closure.

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3.4.13.3.2 Feed and drain connections. - Feed and drain connections shall be of sizes suitable for the purpose intended. The design shall be such that the clean solvent will flow into the shell above the normal solvent level and the dirty solvent can be pumped from the lowest point in the bottom of the shell for circulation through the button trap, pump, and filter.

3.4.13.4 Cylinder. - The cylinder shall be either of the reversing or nonreversing action. The cylinder shall be fabricated of perforated corrosion-resisting steel or of nickel-copper alloy with a minimum of 60 percent nickel and shall be of welded or riveted construction. The cylinder shall be amply reinforced for extraction and shall rotate on heavy duty, anti-friction bearings.

3.4.13.4.1 Lifting ribs. - Cylinders shall be equipped with suitable number of ribs. The ribs may be solid or perforated and shall be of the same material as the cylinder plates.

3.4.13.4.2 Cylinder doors. - Cylinder doors of corrosion-resisting steel or of nickel-copper alloy with a minimum of 60 percent nickel shall be provided on side-loading units. Two-pocket cylinders, where equipped with doors, shall have one cylinder door for each cylinder compartment. Cylinder doors, where used, shall be suitably reinforced and perforated and fitted with rugged latches and keepers for securing door in open or closed positions. Latches and keepers shall be readily renewable and shall be designed to take-up wear automatically, thereby keeping the cylinder door or doors tight at all times when in a closed position. Cylinder doors shall be rigid, fit tight, and shall be suitably reinforced to prevent distortion. Door fittings and guides shall be of corrosion-resisting steel, nickel-copper alloy with a minimum of 60 percent nickel, or other mateial approved by the bureau or agency concerned.

3.4.13.4.3 Cylinder finish. - The inside of the cylinder shall have no rough or sharp projections and shall be smooth on the edges.

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3.4.13.4.4 Seals. - Means shall be provided to prevent the lubricant from entering parts of the dry-cleaning unit where clothes are handled, and prevent dry-cleaning solutions from entering the bearings. Bearing housing and cleaning chamber shall be sealed.

3.4.13.4.5 Inching. - Means shall be provided for bringing the cylinder door to the proper loading position and securing the cylinder in this position.

3.4.13.5 Base storage tank. - The storage tank shall be constructed of heavy steel plate, lead-lined, or coated on the inside with baked-on plastic, or any other suitable coating or lining. The tank shall be fitted with button and lint trap, filler cap, solvent level gauge, and clean-out plate for cleaning out the tank. The tank shall be provided with a cooling coil.

3.4.13.6 Steam-heating coils. - Steam-heating coils shall be of copper and of the extended surface type, suitable for a steam working pressure of 100 p.s.i., and shall be designed to deliver heated air through the material being tumbled in an efficient manner which will evaporate the solvent and remove and dissipate the solvent odors.

3.4.13.7 Filter and still. - The filter and still shall be constructed of suitable ferrous and nonferrous materials and shall be of the sizes and capacities suitable for the purpose intended. The still shall have a capacity of not less than 25 gallons per hour.

3.4.13.7.1 Condenser. - The condenser shall be of the tube type with ample capacity for continuous full-capacity operation. The condenser shall be designed for use with salt-water.

3.4.13.8 Brakes. - Magnetic brakes, if used, shall be in accordance with MIL-B-16392 for Navy class "C" service.

3.4.14 Safety devices. - The equipment shall incorporate adequate safety devices. It shall not be possible to operate the unit with the shell door open. Exposed belts, chains, shafts, pulleys, gears, and other moving parts shall be fully enclosed or guarded so as to prevent contact with same by operators.

3.4.15 Door latches. - Latches for doors shall be of ample size, shall be located so as to permit ready operation, and shall have sufficient clearances to avoid injury to the operator's fingers and hands when doors are being opened or closed.

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3.4.16 Equipment bases. - Bases shall be flat on the bottom and lugs for anchor bolts shall be flush with the bottom of the base or slightly relieved from same so that the dry-cleaning unit will be suitable for installation on a flat surface. Anchor bolt holes of adequate size for the installation shall be provided.

3.4.17 Vibration and impact. - Dry-cleaning units made of material other than malleable iron, ductile iron or steel shall be capable of passing the vibration and impact tests specified in 4.2.3.2, 4.2.3.2.1 and 4.2.3.2.2. The units shall be capable of being operated after the vibration and impact tests. The contractor shall certify in writing that the dry-cleaning units meet one of the following conditions:

(a) Frames, base and strength members are constructed of malleable iron, ductile iron or steel.
(b) Dry-cleaning unit has passed the vibration and impact tests specified herein and is capable of being operated after these tests.

3.5 Lubrication. - Lubrication instructions shall be provided and incorporated in the technical manuals (see 3.11).

3.6 Corrosion protection. - The complete units shall be suitably protected against corrosion and present a neat appearance.

3.7 Color of finish. - Dry-cleaning units shall be painted a basic green color, except on corrosion-resisting metal and except when they are suitably protected by plating. Exposed heated surfaces, such as tumbler heating coil housing, shall be painted with an aluminum paint of heat resisting grade.

3.8 Identification plates. - The name and size (in inches) of the item, the manufacturer's name, trade name or trademark of such known character as to be easily identifiable, the manufacturer's address, serial number, contract number, and year of manufacture, all in plainly legible characters, shall be stamped, etched, or embossed on a corrosion-resisting metal or brass identification plate or plates and securely affixed to each component item of equipment in a conspicuous place. Code or serial numbers will not be accepted as complying with this requirement. Each dry-cleaning unit shall have a plate designating the capacity in pounds, dry weight, per load. In addition, the identification plate shall list the stock number which will be furnished the contractor not later than the time of approval of drawings.

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3.9 Onboard and stock repair parts. - The procedure for the determination, procurement, and identification of onboard and stock repair parts shall be in accordance with appendix II of MIL-P-15137. In completing the information required in NAVSHIPS 4786, furnished in accordance with step 1 of appendix II, the actual "on order" repair parts shall be indicated.

3.9.1 Onboard repair parts for the electrical equipment shall be furnished in accordance with the applicable specification.

3.10 Drawings. - Contractor's drawings for the assembled dry-cleaning unit shall include the following information:

(a) General assembly of the unit, including such cross-sectional or exploded views as are necessary to depict the various operating parts. Electrical components mounted on the unit shall be shown, but variation in outline will be permitted so that components of different manufacture may be used.
(b) Complete overall and mounting dimensions, including size of bolt holes of the units in inches and fractions of an inch.
(c) Salient design data shall be shown, such as weight of units, capacity, speed, washing, extracting, and drying and power of driving motor, pump and fan ratings, filter and still capacities. Mechanical repair parts and special tools shall be listed and designated by names and part numbers.
(d) Wiring and schematic diagrams including description of operation of controllers and related devices.
(e) List of materials used including specification or composition information, source and manufacturer's catalog number.

3.10.1 Approval. - The contractor shall submit drawings for approval to the bureau or agency concerned, via the Government inspector, within 60 days after date of contract or order. The bureau or agency concerned will indicate such changes and modifications as are deemed necessary. It is estimated that it will take 20 days to review the preliminary master drawing. Production equipment shall be manufactured in accordance with preliminary master drawings as approved by the bureau or agency concerned. In event the contractor does not receive notification of approval, conditional approval, or disapproval of working drawings within 20 days after submission thereof, and promptly so notifies the contracting officer in writing, the contract delivery dates for the production equipment shall be extended by the contracting officer by the number of days by which notification is delayed. After approval of the drawings, subsequent approval on future contracts need not be made. The contractor shall certify in writing that the dry-cleaning unit to be furnished is identical with the unit previously furnished and shall provide the contract number of the previous contract.

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3.10.1.1 Any work done prior to approval of preliminary master drawings by the bureau or agency concerned will be at the contractor's own risk.

3.11 Technical manuals. - Technical manuals shall be of a commercial type consisting in general of manufacturer's standard literature and illustrations or drawings specified in 3.10. The manuals shall concisely outline the proper operation, maintenance and parts identification of the units and their controls. The manuals shall be suitable bound in a cover of durable quality. Two manuals shall be packed with each dry-cleaning unit.

3.12 Workmanship. - Each finished component assembly shall be clean, well made, and free from any defects which affect its appearance or which may affect its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. - Prior to the start of production and within 30 days after the date of award of the contract or order, one machine of each design and size shall be submitted to the U.S. Naval Supply Research and Development Facility, Bayonne, New Jersey, for tests of 4.2.3.1 through 4.2.3.2.2. The preproduction sample is defined as a production line model of the product which the manufacturer will subsequently produce in fulfillment of the contract. Approval of the preproduction sample by the bureau or agency concerned shall not relieve the contractor of the responsibility of compliance with the requirements of this specification. Any change or deviations from the preproduction sample shall be submitted for the approval to the bureau or agency concerned. The approved preproduction sample will be used as a guide in the inspection of production machines, and shall be presented for acceptance as the last unit for delivery under the contract.

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4.2.1 Prior to the submission of the preproduction sample for testing, the manufacturer shall subject the machine to the test of 4.3.3. Certification containing actual test data shall be furnished showing that the sample submitted for preproduction tests has successfully passed the test of 4.3.3.

4.2.2 When a contractor has previously supplied equipment on a recent contract or order identical to that covered by this specification which has passed the preproduction tests specified herein, the requirements for preproduction tests may be waived by the bureau or agency provided acceptable test data is furnished.

4.2.3 Preproduction tests.

4.2.3.1 Inclination. - The machine shall be inclined at an angle of 30 degrees to each side of two vertical planes at right angles to each other and shall operate satisfactorily in each of the four positions (see 3.4.2).

4.2.3.2 Vibration and impact tests. - The machine shall be tested without packaging or crating. The vibration test shall be performed prior to the impact test. The machine tested shall be capable of performing its principal functions after testing.

4.2.3.2.1 Vibration test. - The machine shall be placed on the table of a vibration tester. Fences may be attached to the test table suitable for the machine being tested. While the machine is performing its washing cycle only (washing a rated machine capacity load) it shall be vibrated so that it leaves the table momentarily at some interval during the vibration cycle. The proper vibration frequency shall be such that a 1/16 inch thick shim may be inserted between one bottom edge of the machine and the platform of the machine. The machine shall be vibrated for 2 hours. The function of this test table is to produce a measured oscillation upon the equipment to determine its ability to withstand dynamic forces due to vibration which may be encountered aboard ship. The machine shall function satisfactorily during this test.

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4.2.3.2.2 Impact test. - An inclined testing device similar to the Conbur machine or equal equipment producing equivalent results, and a shock recorder shall be used. The machine shall be fastened to the base such that the machine will make no physical contact with the barrier of the impact machine. The assembly shall be placed on the dolly with the face or edge of the base which is to receive the impact projecting approximately 2 inches beyond the forward end of the dolly. The shock recorder shall be positioned on the machine to record the maximum shock received during the impact test. The dolly shall be drawn up the incline to a position which will produce an impact which has a vertical component equivalent to 10g.

4.3 Acceptance tests. - In addition to the tests specified herein physical, chemical, finish and any other tests deemed necessary by the Government shall be conducted to determine that the equipment meets the requirements of this specification.

4.3.1 Operation test. - Each complete unit shall be operated with a full change of solvent for three complete operating cycles to determine satisfactory performance of the unit and its component parts.

4.3.2 Unbalanced load vibration test. - Each dry-cleaning unit shall be secured using the bolt holes provided for anchoring the unit to the deck. With one pocket of the unit loaded with a soaked load of 1/3 of the rated unit capacity of dry laundry and the vibration cut-out switch disconnected, the unit shall be fully operated following the completion of an average extracting cycle. It shall be determined that the vibration limit switch protects. the dry-cleaning unit.

4.3.3 Insulation. - The insulation resistance of each motor, controller and connecting cable shall be measured by a 500 volt d.c. insulation resistance tester. Insulation resistance of each component shall be not less than 10 megohms.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.1.1 Dry-cleaning unit.-

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5.1.1.1 Preservation and packaging. - Preservation and packaging of the dry-cleaning unit and accessories shall be sufficient to afford adequate protection against corrosion, deterioration and physical and mechanical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number, manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.- Onboard repair parts shall be preserved and packaged Level A, packed Level C and marked Levels A and C in accordance with MIL-R-196, MIL-E-17555 or the commodity specification as applicable. Items not covered therein shall be preserved and packaged in accordance with the guide lines of MIL-P-116. Items shall be individually packaged except when used in sets.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be as specified (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging. - The unit shall be preserved and packaged in accordance with the requirements of MIL-P-3153 for extended storage or overseas shipment. Detached electrical controllers shall be preserved and packaged in accordance with Level A requirements of MIL-C-2212 or MIL-C-2174 as applicable.

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5.2.1.1.1 Repair parts. - Repair parts shall be preserved and packaged Level A in accordance with MIL-R-196 and MIL-E-17555 as applicable.

5.2.1.2 Packing. -

5.2.1.2.1 Level A. - The unit shall be packed in accordance with the overseas shipment and extended storage requirements of MIL-P-3153.

5.2.1.2.2 Level B.- The units shall be packed in accordance with the domestic shipment requirements of MIL-P-3153.

5.2.1.3 Repair parts. - Repair parts shall be packed Level A or B in accordance with MIL-R-196, MIL-E-17555 or the commodity specification as applicable.

5.2.1.4 Marking.- In addition to any special marking required, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Motor voltage required (see 3.4.12).
(c) Size and capacity of cylinder (see 3.4.13.2).
(d) Selection of applicable levels of preservation, packaging, and packing if other than as specified in 5.1 (see 5.2).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 3510-N012Sh)

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R399

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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